Ensembling learning techniques

1) bagging — Classification – best vote decision
2) boosting — regression – average of decision

Random Forest = with parallel crossing problem

no. of decision trees } parameters

Heavy computation & Heavy loss

Boosting techniques

Ada Boost (adaptive boosting) → manually apply weights

gradient Boosting → apply gradient descent to apply weights

Extreme gradient boosting

cat Boosting

Ligh GBm

Imbalance data

→ high variance in data

XGboost – regularise data better than normal gradient boost trees

extreme gradient boosting

– handles missing values in data (00 sparse data

– It adds double derivate.

– Internally apply cross-validation techniques

→ It is used for categorical data

⇒ Catboost - Categorical boosting

→ Internally handles the categorical data,

→ take this type of data & converts to numerical

⇒ Light GBm

L purely works for high volume of data.

KNN → nearest + distance

L eucledian distance - diff x & y coordi

✓ Supervised learning

L manhattan distance.

L

K nearest neighbour (KNN)

Svm (support vector machine)

· margin

support vectors

hyper plane